

karP: an experiential prototype for Kinesthetic Augmented Reality on mobile — Playful, Portable, Producible

Zhaohan Pan
Stanford University
Stanford, California, USA
yoas0b1@stanford.edu

Zhihao Meng
University of California, Los Angeles
Los Angeles, California, USA
zhm@ucla.edu

Gaolin Ge
University of Washington
Seattle, Washington, USA
gge2@uw.edu

Hongrui Wu
University of Washington
Seattle, Washington, USA
hrwu@uw.edu

Haoran Lu
University of Washington
Seattle, Washington, USA
hrlu24@uw.edu

Qifeng Yang
University of Washington
Seattle, Washington, USA
qyang24@uw.edu

Jingyang Liang
Aiglon College
Ollon, Vaud, Switzerland
liajin20@aiglon.ch

Abstract

Kinesthetic interaction systems on mobile augmented reality (MAR) platforms remain largely conceptual. To explore this design space, we present karP, an experiential prototype that serves as a proof of concept for integrating kinesthetic systems with MAR. karP's device features a manipulandum-exoskeleton design that supports kinesthetic input and output for striking gestures with handheld items. Leveraging the native capabilities of MAR, the smartphone functions simultaneously as the visual-spatial interface and spatial tracker, coordinating with the kinesthetic device to control the avatar's physical interaction in virtual environments. We further introduce game-mechanism-inspired experiential scenes to demonstrate diverse interactions made possible by our system. Ultimately, karP embodies our vision for kinesthetic design in MAR that is playful, portable, and producible.

CCS Concepts

• **Human-centered computing** → **Haptic devices; Mixed / augmented reality.**

Keywords

Kinesthetic device, Mobile augmented reality

ACM Reference Format:

Zhaohan Pan, Gaolin Ge, Haoran Lu, Zhihao Meng, Hongrui Wu, Qifeng Yang, and Jingyang Liang. 2025. karP: an experiential prototype for Kinesthetic Augmented Reality on mobile — Playful, Portable, Producible. In

Designing Interactive Systems Conference (DIS '25 Companion), July 05–09, 2025, Funchal, Portugal. ACM, New York, NY, USA, 5 pages. <https://doi.org/10.1145/3715668.3735607>

1 Introduction

Device-based haptic systems enable embodied interaction in Extended Reality (XR) environments [12]. While such systems have seen significant progress in Virtual Reality (VR), where mature hardware and software infrastructures support complex implementations, development in Mobile Augmented Reality (MAR) remains limited: MAR's portability leads to constraints such as limited field of view, lack of dedicated spatial tracking hardware, and the need for manual phone manipulation [1]. These factors make it challenging to integrate high-fidelity haptics into MAR, particularly in interactions requiring additional device manipulation and rendering. Nonetheless, MAR's scalability, accessibility, and similar capability of spatial computing make it a promising platform for expanding XR haptics beyond traditional VR contexts.

One prior work demonstrates that smartphones can serve simultaneously as the visual interface and spatial tracker in a MAR-based haptic system [13]. This design effectively leverages the inherent functionalities of MAR to reduce hardware requirements. However, the haptic pattern in its design is relatively simple and focused only on localized cutaneous feedback at the fingertip, which makes such a lightweight setup feasible. In contrast, as a more expressive and playful XR interactions, the kinesthetic systems typically demand a more complex mechanical setup and increased hand involvement. A separate work explores the use of Leap Motion to capture dynamic hand gesture input [14], while this method is not suitable for in-device manipulation gestures (where the device's structure can obstruct hand tracking) and also requires additional hardware. The exploration of the MAR-kinesthetics area is still constrained by limited resources, and the integration of wearable mechanical devices into MAR platforms remains at a conceptual stage [1]. Inspired by the prior work's implementation of MAR haptics, we aim to explore how a similar design strategy could scale to support more physically engaging and playful kinesthetic scenes.

Permission to make digital or hard copies of all or part of this work for personal or classroom use is granted without fee provided that copies are not made or distributed for profit or commercial advantage and that copies bear this notice and the full citation on the first page. Copyrights for components of this work owned by others than the author(s) must be honored. Abstracting with credit is permitted. To copy otherwise, or republish, to post on servers or to redistribute to lists, requires prior specific permission and/or a fee. Request permissions from permissions@acm.org.

DIS '25 Companion, Funchal, Portugal

© 2025 Copyright held by the owner/author(s). Publication rights licensed to ACM.

ACM ISBN 979-8-4007-1486-3/25/07

<https://doi.org/10.1145/3715668.3735607>

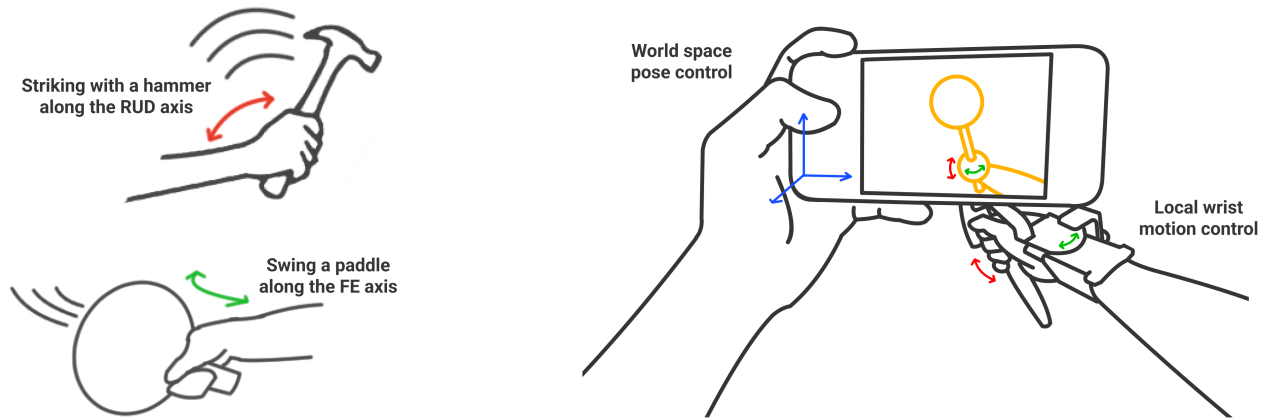


Figure 1: Demonstration of featured gestures and control method of karP. Left: example item-holding impulse gestures. Right: decomposed control on avatar using the phone and the kinesthetic device for AR interaction.

To support this goal, we present **karP**, a MAR-kinesthetic experiential prototype composed of a custom wearable device and an iOS application. We envision our contributions as follows:

- A proof of concept that demonstrates kinesthetic interaction within a MAR context.
- A complete experiential prototype featuring a low-cost, reproducible hardware design and a MAR interface developed using mainstream frameworks.
- A reflection on the design logic and system architecture that may inform future research and applications in this space.

2 Kinesthetic Gesture and MAR System

Designing kinesthetic devices typically requires tailoring the mechanical response to specific interaction motions, such as grasping [3, 15] or encounter-based [9] gestures. In this work, we explore item-holding impulse gestures as a particularly playful example. These gestures are widely observed in XR interaction scenarios, especially in gaming contexts such as paddle-based hitting, sword slashing, and tool striking [4, 7]. Analyzing these interactions, we find that they primarily involve two main anatomical axes of the human wrist: Radial/Ulnar Deviation (RUD) and Flexion/Extension (FE) [2]. Based on this simplification, our design focuses on supporting these two Degrees of Freedom (DoF) with examples shown in **Figure 1 left**, and explores a less commonly emphasized manipulandum-exoskeleton format that balances playfulness with movability in MAR. We expect users to manipulate an avatar within the AR scene through the kinesthetic device and to feel virtual haptic feedback on the avatar.

To build a MAR-based kinesthetic system, we revisit four essential components: the physics engine, the visual-spatial interface, spatial tracking, and the kinesthetic device. We adopt a similar concept to the previously mentioned design [13], which utilizes the MAR platform as a multifunctional tool that integrates physics, visualization, and tracking. Their approach principally decomposes gesture tracking into two layers: using the smartphone as a world space tracker and relying on IMU modules for local finger pose

estimation. Since their featured gesture involves only simple pressing motions, the system allows the phone to be mounted on the forearm for a reasonable top-down view of the screen. However, our targeted interaction involves more complex and expressive wrist gestures, which people naturally tend to observe from a broader field of view and varied angles, making their forearm-mounted setup less suitable in this context. Therefore, we adopt the idea of decomposed control but reconfigure the setup: the phone is held in one hand to control the view and avatar's world position and orientation, while the kinesthetic device held by the other hand captures localized wrist motions (**Figure 1 right**). This configuration maintains the most conventional form of MAR interaction while enabling the visual perspective and input flexibility required for more complex gestures. Although the bimanual setup may initially seem unintuitive, our preliminary evaluation suggests that it is both feasible and usable. This is also why we share this design as a reproducible prototype, inviting the community to engage with and evaluate our early exploration of kinesthetic interaction in MAR.

3 Wearable Manipulandum Device

karP's kinesthetic device is shown in **Figure 2**. It is designed to be worn on the forearm in an exoskeleton-like configuration, allowing reaction forces during kinesthetic feedback to be supported directly by the user's body. The core manipulandum employs a two-joint spherical mechanism, whose mechanical structure is responsible both for capturing user input and delivering impulse haptic feedback. We specifically aligned the two joints with the user's wrist axes, which makes the two DoF independent. This alignment allows us to directly define the workspace based on the human wrist's natural range of motion and to compute impulse torques with respect to the wrist. The final workspace spans 15° RD, 40° UD, 80° F, and 30° E, fully covering the typical range engaged in our representative gestures [8]. For actuation, we implemented the classic capstan structure (often found in kinesthetic manipulanda [5, 10]), which amplifies the relatively weak torque output from compact brushless DC motors while ensuring smooth transmission during manipulation [6]. In our setup, maximum torque

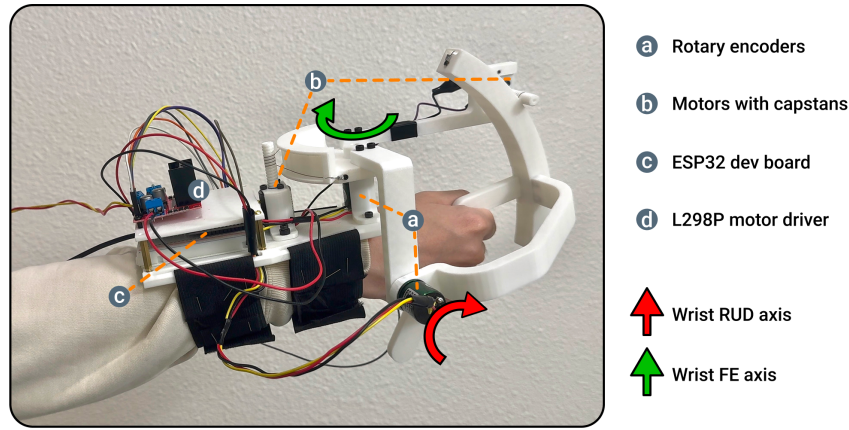


Figure 2: karP’s manipulandum-exoskeleton kinesthetic device. All discussed components and rotary DoF are labeled accordingly.

output is 1.16 N-m on the RUD axis and 0.70 N-m on the FE axis. All structural components are fabricated via 3D printing, assembled with screws, and connected using metal cables. In fact, the device design builds upon the open-source educational resource Hapkit [11], allowing the aforementioned design principles to be explored through their publicly available materials.

The device is controlled by an ESP32 development board, which establishes the interaction service responsible for transmitting user-performed gestures, receiving rendered feedback data, and driving actuation accordingly. It interfaces with an L298P motor driver module to control motors on both DoF. The device incorporates two 2225CL-12 motors and two P3022-V1-360 rotary encoders; the encoders are directly mounted at the joints, serving both as rotational bearings and as sensors for capturing wrist rotation. The motors are connected via modular adapters, allowing users to easily replace or experiment with different motor configurations. All the components used are considered basic and highly accessible electromechanical parts, consistent with our vision of building an accessible and reproducible experiential prototype.

4 MAR Setup and Haptic Physics

We developed karP’s AR application for iOS to run on iPhones, leveraging SceneKit for scene construction, physics simulation, and kinesthetic feedback computation, and ARKit for device camera tracking, environmental understanding, and spatial frame stabilization.

To enable seamless space transformations of the avatar through manual manipulation of the phone, we configure it as a kinematic object with its wrist anchored to the AR camera. All other physics objects in the scene are configured as dynamic bodies to support physical interactions with the avatar. Since SceneKit does not calculate collision impulses for kinematic bodies, we implemented a customized physics model to manually estimate interaction forces. We assign spherical collision bounds to all dynamic objects and disregard shear force components. The normal force is then approximated based on the object’s moving direction and acceleration at the moment of contact, independent of the avatar’s geometry or

striking angle. As the featured gestures primarily involve object surface-normal contact, our simplified estimation provides a reasonable approximation of impulse events. Since mainstream MAR toolkits lack native support for detailed force computation (especially when targeting kinesthetic feedback), our lightweight custom model serves as a necessary adaptation rather than an optimization, ensuring the core interaction remains feasible within existing development environments.

We assume a rigid connection between the user’s hand and the held item (i.e., the user holds perfectly firm), allowing torque around the wrist to be computed by combining the reaction force vector with the lever arm extending from the wrist to the contact point. Both the reaction force and the contact point are transformed into the avatar’s local space, and the resulting torque is then projected onto the RUD and FE axes, respectively. This processed impulse feedback is finally sent back to the server for machine output.

5 Interactive Scenes

Inspired by the interaction mechanisms found in XR games, we designed a *Ball Playground* for karP, as its interfaces and some in-scene interaction demonstrations shown in **Figure 3**. We introduce variation in the initial states of objects (static, dynamic), their physical properties (mass, radius), and the avatar’s holding items, where all are unified by the core goal of engaging users through diverse and expressive physical interactions around the featured gesture.

We began by sampling object properties based on three familiar reference models: ping pong balls ($r = 0.02$ m, $m = 0.0027$ kg), tennis balls (0.0335 m, 0.057 kg), and basketballs (0.12 m, 0.62 kg). These are randomly generated within scenes to provide perceptual variety and simultaneously give users a real reference. Additionally, we designed two interactive items for the avatar: a lightsaber and a ping pong paddle, each offering a different visual and geometric interface. As discussed previously, the items’ geometry is decoupled from physical collision and impulse calculation, enabling seamless extension to additional designs in the future.

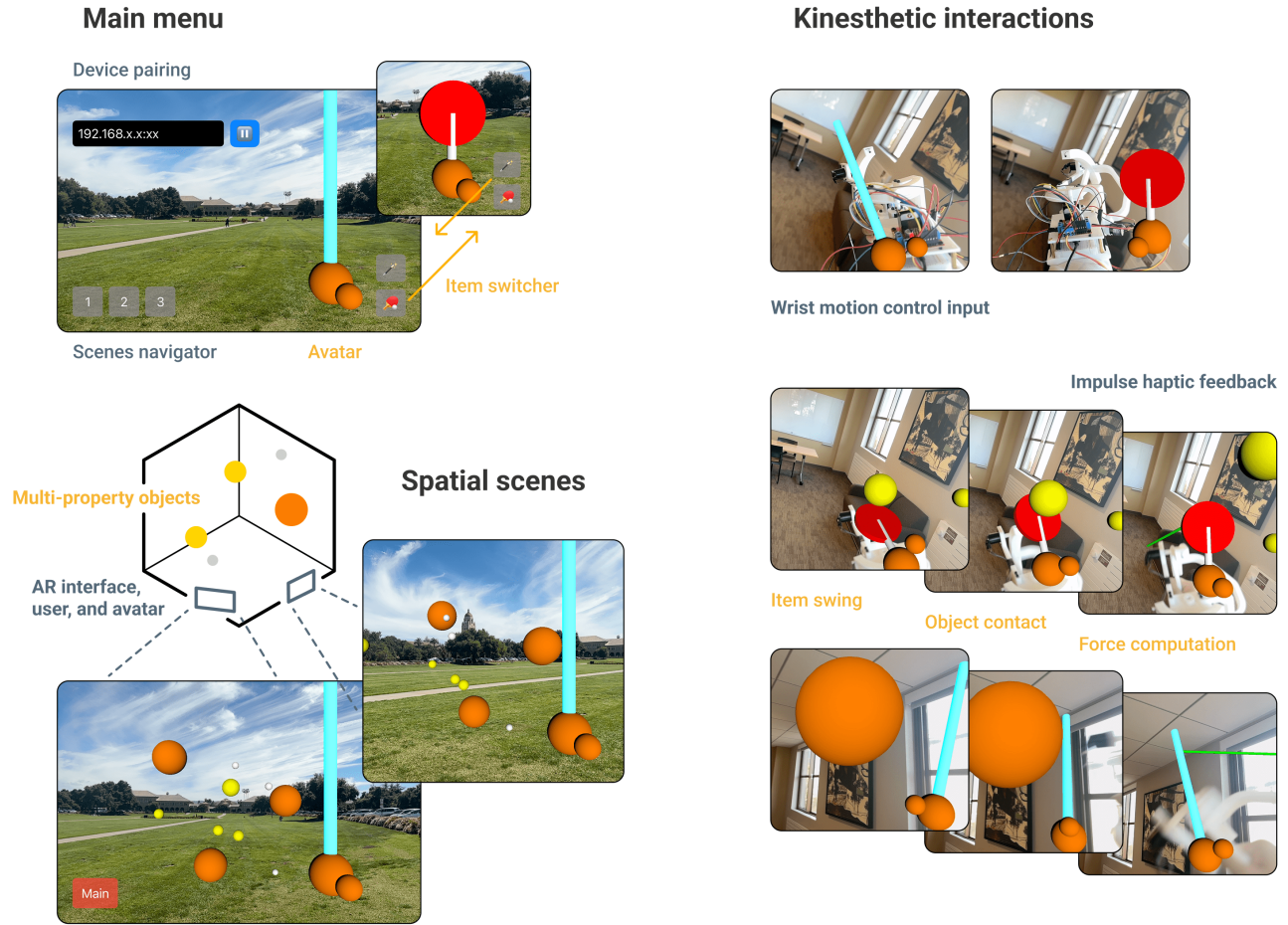


Figure 3: Demonstration of karP's Ball Playground. Main menu: interface for App logistics. Spatial scenes: scenes for user to interact with dynamic objects. Kinesthetic interactions: device-enabled kinesthetic input / output. Green line visualization of impulse force shown here.

We implemented three interactive scenes to feature these gestures. *Frozen Balls* places static balls randomly around the user, allowing them to proactively initiate contact and experience the response of objects with varying physical attributes. *Serve Machine* presents a launcher-like setup where balls are spawned at a distance and projected toward the user at varying velocities. This enables both reactive blocking and active striking, similar to a real-world ball machine. Finally, *Ball Ninja*, inspired by *Fruit Ninja*, launches balls vertically from below under gravity, creating an arcade-style slashing experience. We also deploy a dynamic green line to give visualization to the computed impulse force during object contact, helping users better perceive the force feedback during the interaction.

6 Conclusion and Future Works

karP demonstrated our preliminary exploration of integrating a complete and complex mechanical kinesthetic input / output system into a MAR platform. Our design achieved an initial level of playfulness while adapting to the inherent interaction style of MAR.

As for the haptic device, we realized the vision proposed in prior studies by successfully deploying a wearable manipulandum. We also kept both the hardware and software built with accessible resources and mainstream development frameworks, so as to echo the scalability of MAR.

For future work, the decomposed control design still requires further user evaluation. It is also possible to incorporate additional hardware and lightweight tracking algorithms to relax current constraints, such as the rigid wrist anchoring or perfectly firm item holding. This direction, in fact, leads to a consideration on evaluating platform suitability: the extent of added infrastructure must remain compatible with the lightweight and portable nature of MAR. Therefore, understanding where this threshold lies becomes an important consideration, which can inform future decisions about how to best design haptic systems for MAR.

Acknowledgments

We thank Danyang Fan for reviewing this project and also Shilong Qiu's collaboration in producing the visual materials.

References

- [1] Carlos Bermejo and Pan Hui. 2021. A Survey on Haptic Technologies for Mobile Augmented Reality. *Comput. Surveys* 54, 9, Article 184 (oct 2021). <https://doi.org/10.1145/3465396>
- [2] Bethany U. Casagrande (Ed.). 2022. *MRI of the Upper Extremity: Elbow, Wrist, and Hand*. Springer. <https://doi.org/10.1007/978-3-030-81612-4>
- [3] Inrak Choi and Sean Follmer. 2016. Wolverine: A Wearable Haptic Interface for Grasping in VR. In *Adjunct Proceedings of the 29th Annual ACM Symposium on User Interface Software and Technology (UIST '16 Adjunct)* (Tokyo, Japan). Association for Computing Machinery, New York, NY, USA, 117–119. <https://doi.org/10.1145/2984751.2985725>
- [4] Javier Dehesa, Andrew Vidler, Christof Lutteroth, and Julian Padget. 2020. Touché: Data-Driven Interactive Sword Fighting in Virtual Reality. In *Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems (CHI '20)* (Honolulu, HI, USA). Association for Computing Machinery, New York, NY, USA, 1–14. <https://doi.org/10.1145/3313831.3376714>
- [5] Jonas Forsslund, Michael Yip, and Eva-Lotta Sallnäs. 2015. WoodenHaptics: A Starting Kit for Crafting Force-Reflecting Spatial Haptic Devices. In *Proceedings of the Ninth International Conference on Tangible, Embedded, and Embodied Interaction (TEI '15)* (Stanford, California, USA). Association for Computing Machinery, New York, NY, USA, 133–140. <https://doi.org/10.1145/2677199.2680595>
- [6] Vincent Hayward and Karon E. MacLean. 2007. Do It Yourself Haptics: Part I. *IEEE Robotics & Automation Magazine* 14, 4 (2007), 88–104. <https://doi.org/10.1109/MRA.2007.907921>
- [7] Eren Karatas, Kissinger Sunday, Sude Erva Apak, Yiwei Li, Junwei Sun, Anil Ufuk Batmaz, and Mayra Donaji Barrera Machuca. 2023. "I Consider VR Table Tennis to Be My Secret Weapon!": An Analysis of the VR Table Tennis Players' Experiences Outside the Lab. In *Proceedings of the 2023 ACM Symposium on Spatial User Interaction (SUI '23)* (Sydney, NSW, Australia). Association for Computing Machinery, New York, NY, USA, Article 29, 12 pages. <https://doi.org/10.1145/3607822.3614539>
- [8] Kwang Gi Kim, Chan Soo Park, Suk Ha Jeon, Eui Yub Jung, Jiyun Ha, and Sanglim Lee. 2018. Feasibility of a New Desktop Motion Analysis System with a Video Game Console for Assessing Various Three-Dimensional Wrist Motions. *Clinics in Orthopedic Surgery* 10, 4 (2018), 468–478. <https://doi.org/10.4055/cios.2018.10.4.468>
- [9] Robert Kovacs, Eyal Ofek, Mar Gonzalez Franco, Alexa Fay Siu, Sebastian Marwecki, Christian Holz, and Mike Sinclair. 2020. Haptic PIVOT: On-Demand Handhelds in VR. In *Proceedings of the 33rd Annual ACM Symposium on User Interface Software and Technology (UIST '20)* (Virtual Event, USA). Association for Computing Machinery, New York, NY, USA, 1046–1059. <https://doi.org/10.1145/3379337.3415854>
- [10] Melisa Orta Martinez, Joseph Campion, Tara Gholami, Michal K. Rittikaidachar, Aaron C. Barron, and Allison M. Okamura. 2017. Open Source, Modular, Customizable, 3-D Printed Kinesthetic Haptic Devices. In *2017 IEEE World Haptics Conference (WHC)*. 142–147. <https://doi.org/10.1109/WHC.2017.7989891>
- [11] Tania K. Morimoto, Paulo Blikstein, and Allison M. Okamura. 2014. [D81] Hapkit: An Open-Hardware Haptic Device for Online Education. In *2014 IEEE Haptics Symposium (HAPTICS)*. 1–1. <https://doi.org/10.1109/HAPTICS.2014.6775560>
- [12] Miguel A. Otaduy, Allison Okamura, and Sriram Subramanian. 2016. Haptic Technologies for Direct Touch in Virtual Reality. In *ACM SIGGRAPH 2016 Courses (SIGGRAPH '16)*. Association for Computing Machinery, New York, NY, USA. <https://doi.org/10.1145/2897826.2927307>
- [13] Gianluca Paolucci, Tommaso Lisini Baldi, Davide Barcelli, and Domenico Praticchizzo. 2020. Combining Wristband Display and Wearable Haptics for Augmented Reality. In *2020 IEEE Conference on Virtual Reality and 3D User Interfaces Abstracts and Workshops (VRW)*. 632–633. <https://doi.org/10.1109/VRW50115.2020.00167>
- [14] Jing Qian, Jiaju Ma, Xiangyu Li, Benjamin Attal, Haoming Lai, James Tompkin, John F. Hughes, and Jeff Huang. 2019. Portal-ble: Intuitive Free-hand Manipulation in Unbounded Smartphone-based Augmented Reality. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology (UIST '19)* (New Orleans, LA, USA). Association for Computing Machinery, New York, NY, USA, 133–145. <https://doi.org/10.1145/3332165.3347904>
- [15] Mike Sinclair, Eyal Ofek, Mar Gonzalez-Franco, and Christian Holz. 2019. CapstanCrunch: A Haptic VR Controller with User-supplied Force Feedback. In *Proceedings of the 32nd Annual ACM Symposium on User Interface Software and Technology (UIST '19)* (New Orleans, LA, USA). Association for Computing Machinery, New York, NY, USA, 815–829. <https://doi.org/10.1145/3332165.3347891>